Do Now · retrieval

Lesson 061 · Thinking Ahead: Inputs, Preconditions and Reuse

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an abstraction assumption?

Q2

Why validate inputs before processing?

Q3

What is a reusable component?

Q4

Array a starts as [2,3,5,7]. Execute a[1]=a[0]+a[2], then a[2]=a[1]-a[3]. Trace the final array, using zero-based indexing.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — Specify the inputs and outputs before choosing algorithms: ticket count and unit price determine total charge.

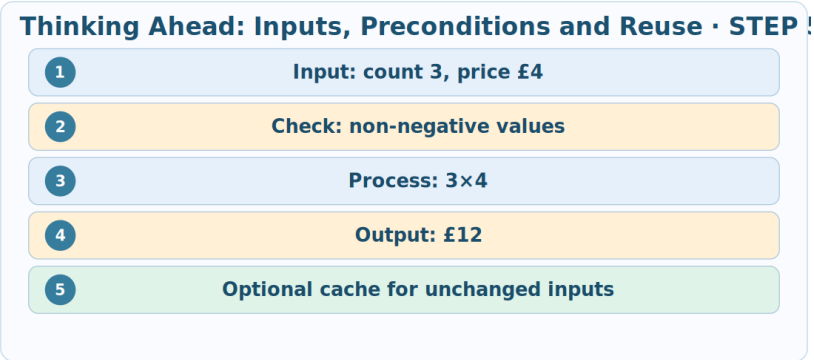
Core idea 2

Model — Make preconditions explicit. Invalid negatives or overflow can produce incorrect outputs; validation and appropriate types reduce risks.

Core idea 3

Evaluate — Caching saves a previously computed result for reuse, trading memory and stale-result risk against repeated calculation time. Reusable methods reduce repeated code.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

precondition — Requirement for valid use.

postcondition — Expected condition after execution.

cache — Saved result for quicker reuse.

module — Reusable program component.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

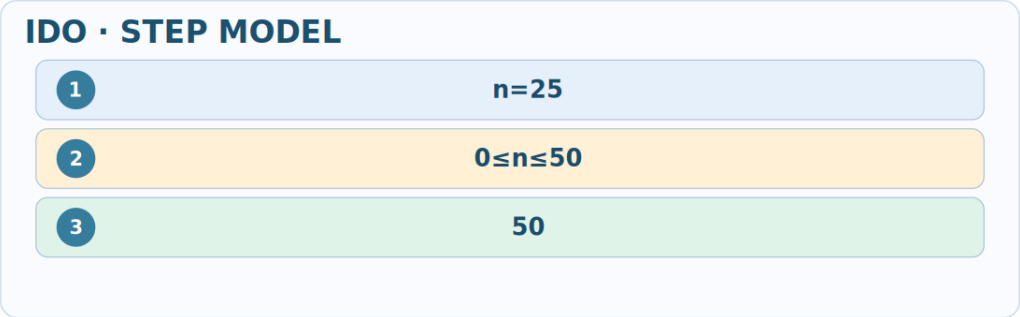
Problem

A function doubles an integer. State input/output and a precondition if it accepts values 0–50 only.

Follow the worked method

1. Explain the decision — The algorithm's result can be correct mathematically but invalid under the declared application rules.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the decision
Explain caching's time/network benefit and correctness trade-off.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For function TicketCost(count,price), state two inputs, output and one precondition. [5]

Question 2

Give a scenario where caching a result saves time, plus a stale-data risk and one invalidation strategy. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

An API-dependent school timetable app calls same room lookup 200 times. Design a reusable method with cache and describe two tests. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Caching a result can never affect correctness.

Use key vocabulary from memory

1. Explain precondition in your own words.

2. Explain postcondition in your own words.

3. Explain cache in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.